AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 35 · STUDENT LESSON

Capstone Train, Analyze, Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One change. Every outcome. Freeze before the final key.

Train and compare a complete capstone revision with an honest evidence boundary.

Keep one habit: Complete kit comparison → pictured error trade-off → runnable candidate trial → freeze ledger → independent separate matrix.

MAIN LESSON · SECTION 1

Can one planned change help—and what still fails?

> Train, compare, explain and freeze before asking for the final answers.

Current capstone goals

I can train and compare reproducible candidates on fixed development data.

I can explain every gain and regression before freezing the selected version.

I can keep final evaluation and any later revision honestly versioned.

Recall reproducibility and source separation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.
- What changes after final-test labels inform a revision?

End goal

By the end, I can explain capstone train, analyze, improve using a traced example and a stated limitation.

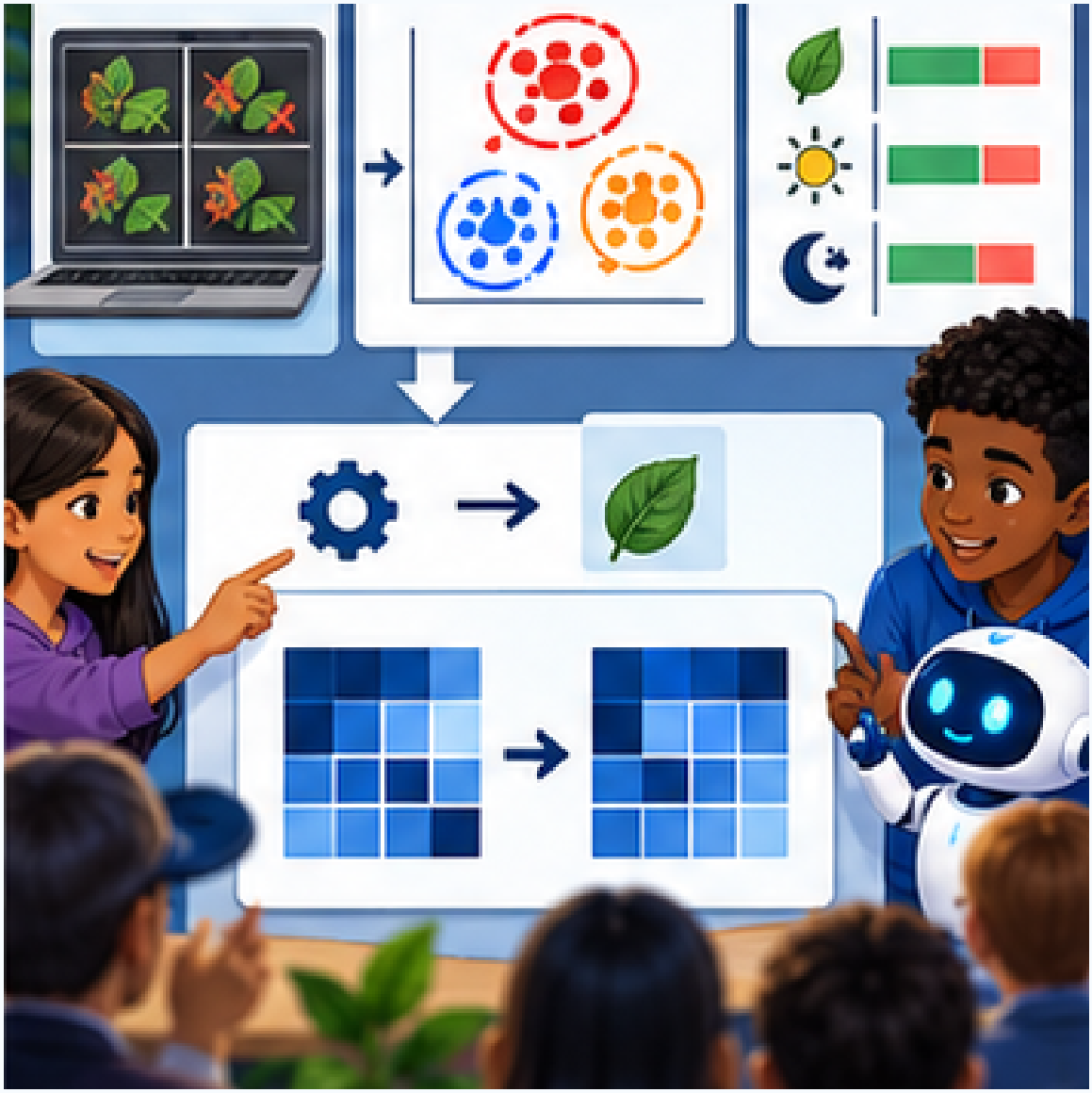
▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.



LEARNING TARGET A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Start with a hypothesis and a fair comparison

> One feature-set change makes the before-and-after result inspectable.

Complete experiment and stage boundaries

Use your teacher-reviewed Plant Classifier project if its data and labels are ready. Otherwise this complete fictional parcel kit is a reproducible practice capstone, not a plant diagnosis. V1 uses width alone. Our planned V2 change adds height because width alone may confuse differently shaped parcels. Keep training rows, validation rows, k=1, distance and tie rules fixed.

Fit feature ranges from training only; compare all 20 validation outcomes, then freeze. Final labels remain blank in student code. Do not mix these results with the separate matrices below.

Final-release rule and fictional scope

The teacher releases final-test labels only after checking the freeze record. Score baseline and candidate on the same planned rows. Save every **prediction**, denominator and failure.

Do not edit a model while reading labels and then present only the edited version as the original test.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

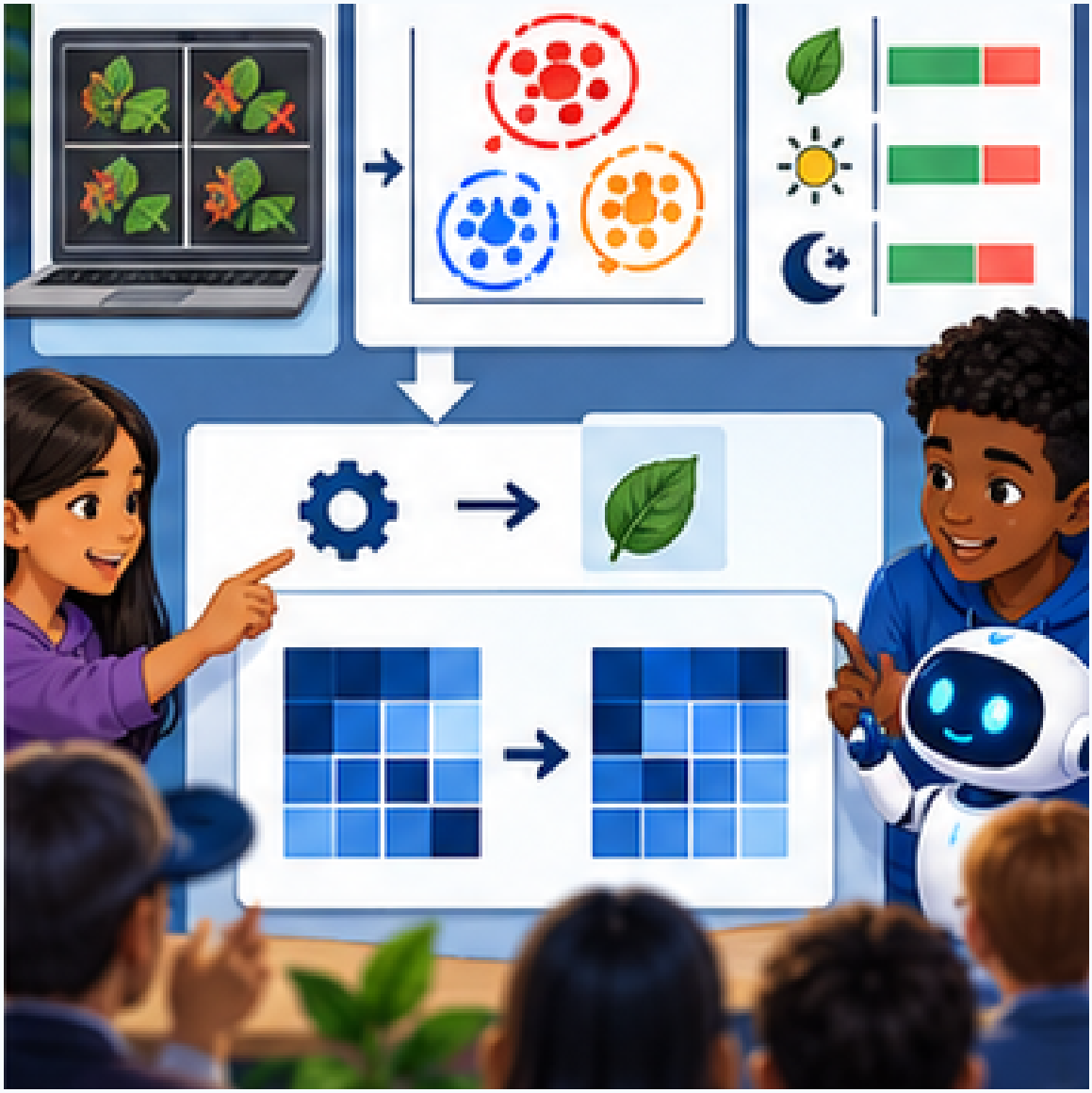
Earlier weeks supplied the model-building **process**; this week examines capstone evaluation and revision.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use capstone train, analyze, improve to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Read the complete comparison before the headline

> Every score names its cases,positive class and model version.

Declared training and comparison contract

The complete program repeats all 72 kit records. Each ID/group is one distinct fictional parcel. Training/validation/final roles are 40/20/12 and stay disjoint. V1 uses width_cm; V2 uses width_cm and height_cm. Scale each feature with its training minimum/range:width 2/8, height 2/7. Use Manhattan distance, k=1, and smallest training ID on an exact distance tie.

Fractions preserve exact arithmetic for these integer dimensions. IDs break ties only; they are not distance features. Choose most validation-correct, then fewer features, then version name. Declare this before reading results.

Complete validation results

Same20 validation parcels; blocked positive · TN · FP · FN · TP · Correct /20 · Blocked found /11

> **Same20 validation parcels; blocked positive:** Training-majority clear baseline
>
> **TN:** 9
>
> **FP:** 0
>
> **FN:** 11
>
> **TP:** 0
>
> **Correct /20:** 9 =45%
>
> **Blocked found /11:** 0

> **Same20 validation parcels; blocked positive:** V1 width only
>
> **TN:** 9

FP: 0

FN: 7

TP: 4

Correct /20: 13 =65%

Blocked found /11: 4

Same20 validation parcels; blocked positive: V2 width and height

TN: 9

FP: 0

FN: 3

TP: 8

Correct /20: 17 =85%

Blocked found /11: 8

Every gain and remaining error

Case-level comparison	Complete saved record
V1 validation mistakes	P24, P42, P49, P56, P60, P64, P67
V2 fixes	P42, P49, P60, P67; four more correct, +20 percentage points
V2 remaining mistakes	P24, P56, P64; all are missed blocked cases
V2 new mistakes among V1-correct cases	None on these 20 inspected validation records
Training accuracy, not final evidence	Baseline22/40, V1 30/40, V2 40/40; V2 can retrieve its own unique training feature pair.

Case-level comparison	Complete saved record
Conclusion and limit	Select V2 under the declared validation rule. These are constructed, inspected development cases; final performance is still withheld.

▼ Separate illustrative final matrix;not the kit result

Illustrative frozen test has 12 rows: candidate[[7, 1],[2, 2]], baseline[[8, 0],[4, 0]], blocked positive. Candidate accuracy 9/12=75 percent, precision 2/3 and recall 2/4; baseline accuracy 8/12 about 66.7 percent.

The candidate adds one false alarm and misses two rather than four blocked cases.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

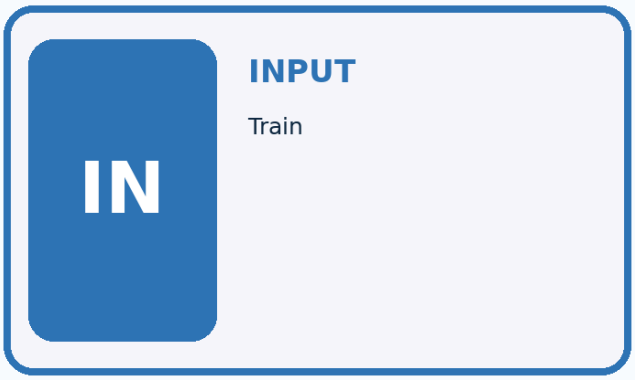
Keep the distinction between an observed result and a proposed next step.

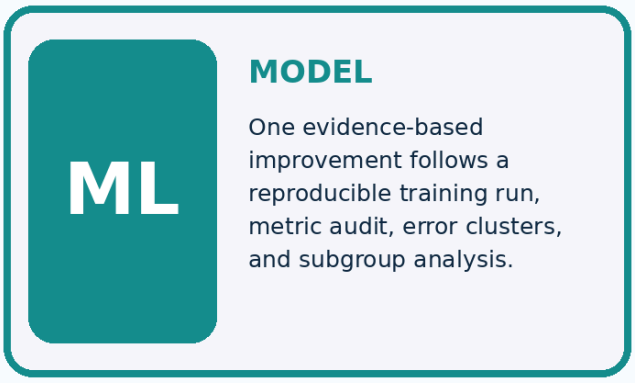
▼ Earlier wording, alternatives and picture transcripts

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Train, audit, change one factor, and compare against the same validation set.

MODEL IDEA One **evidence**-based improvement follows a reproducible **training** run, metric audit, error clusters, and subgroup analysis.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model v1 passes 14/20 validation cases; evening accuracy is 40%. The six failed cases are low-light cases.

One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

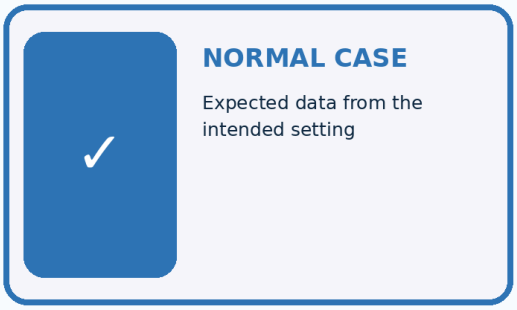
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

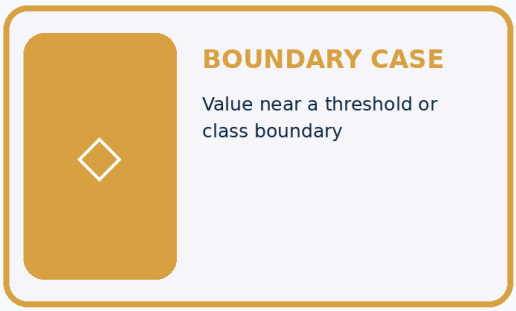
Compare controlled development with an unsupported final claim

> A better validation result still needs a saved freeze record and a distinct final check.

Four evaluation conditions



NORMAL Expected data and expected behavior.

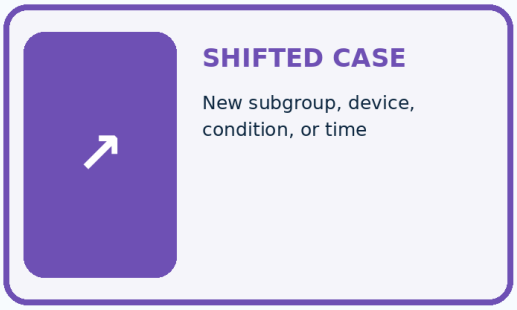


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A revision fixes one miss but creates two extra false alarms.

▼ **Check after predicting**

Prediction: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

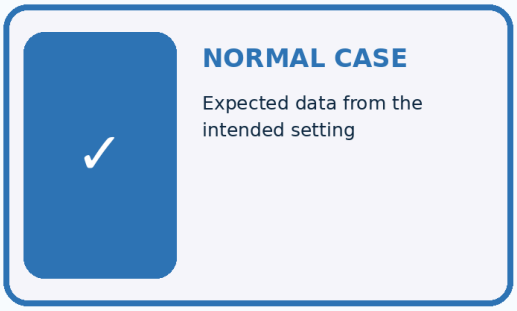
Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

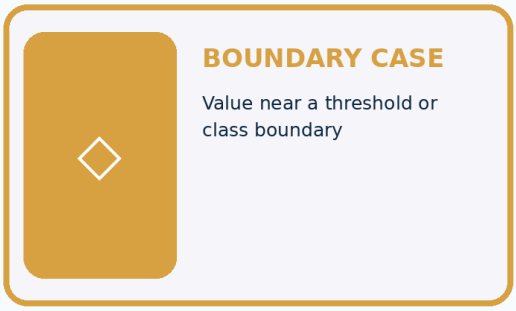
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

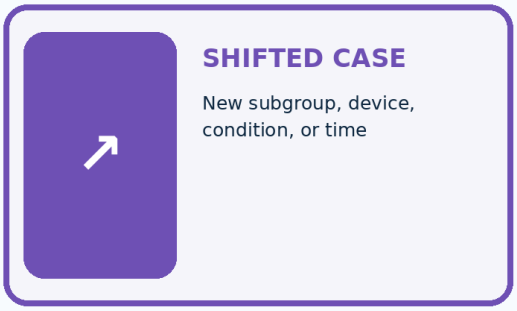
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

A repair can gain one kind of success and lose another

> Read both misses and false alarms instead of naming every revision better.

Separate pictured facts and full arithmetic

Choose the next capstone revision from explicit false alarms and misses.

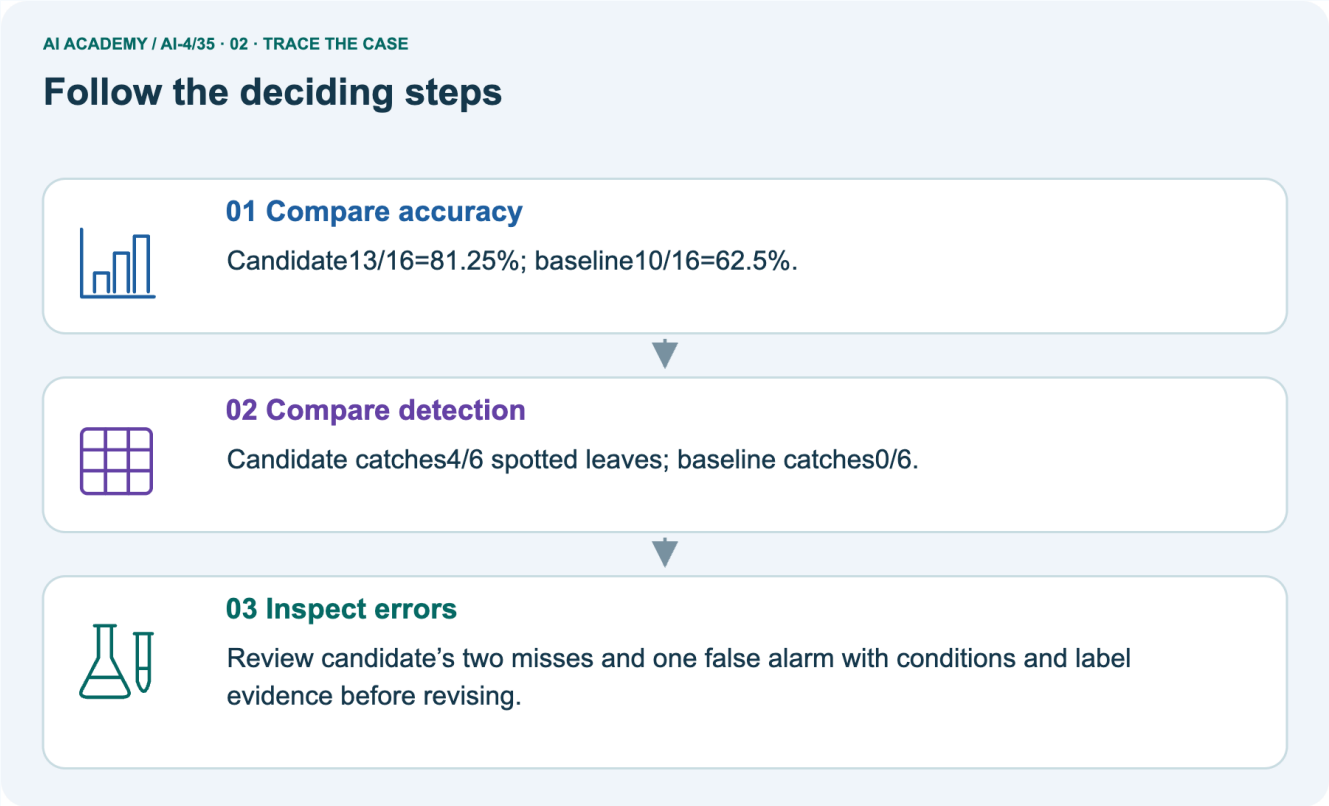
Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate 16-case spotted-leaf arithmetic exercise, not the 72-row parcel kit:original TP 4, FN 2, FP 1, TN 9 gives 13/16 correct and 4/6 recall. The pictured revision fixes one miss but creates two false alarms:TP 5, FN 1, FP 3, TN 7 gives 12/16 correct and 5/6 recall. Higher recall and lower accuracy can coexist.

Which error matters depends on a declared task policy; these 16 cases become development evidence for a revision chosen from their errors.

Trace the original error types



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare accuracy	Candidate13/16=81.25%; baseline10/16=62.5%.
2. Compare detection	Candidate catches4/6 spotted leaves; baseline catches0/6.
3. Inspect errors	Review candidate's two misses and one false alarm with conditions and label evidence before revising.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

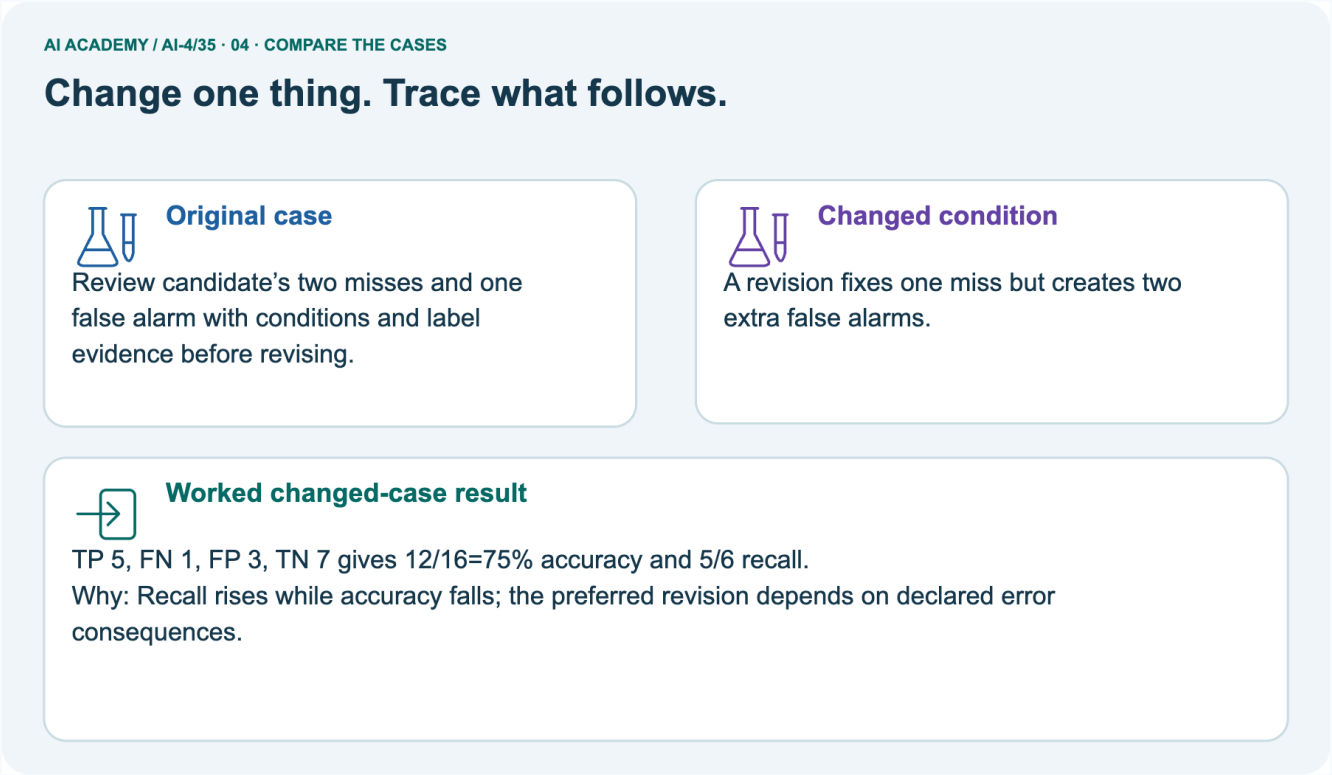
Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the changed trade-off

- Change one thing: A revision fixes one miss but creates two extra false alarms. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain evidence reuse and limits

- Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

- Read the labelled picture: Why is "the revision is better" incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Freeze and release sequence

Capstone evaluation and revision at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Frozen version	Saved before labels	Candidate v 1
First test	9 correct of 12	Report full error table
Proposed revision	Uses observed errors	Needs fresh evaluation

Compare the result with validation without claiming they should be identical. Check error types and coverage. A one-case accuracy difference on twelve rows supports a cautious classroom comparison, not a general reliability guarantee. Keep the original prediction ledger unchanged.

What does the one-correct-case improvement establish? Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 35: Capstone Train, Analyze, Improve

STEP	EVIDENCE AND REASONING
1	Model v1 passes 14/20 validation cases; evening accuracy is 40%.
2	The six failed cases are low-light cases. One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.
3	Add reviewed low-light training examples while freezing features and model settings.
4	Model v2 passes 17/20; evening rises to 70%; inspect every old pass and record remaining limits.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 Model v1 passes 14/20 validation cases; evening accuracy is 40%.

STEP 2 The six failed cases are low-light cases. One of those same six also has a disputed reference label; that is an overlapping review flag, not a seventh failed case.

STEP 3 Add reviewed low-light **training** examples while freezing features and model settings.

STEP 4 Model v2 passes 17/20; evening rises to **70%**; inspect every old pass and record remaining limits.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

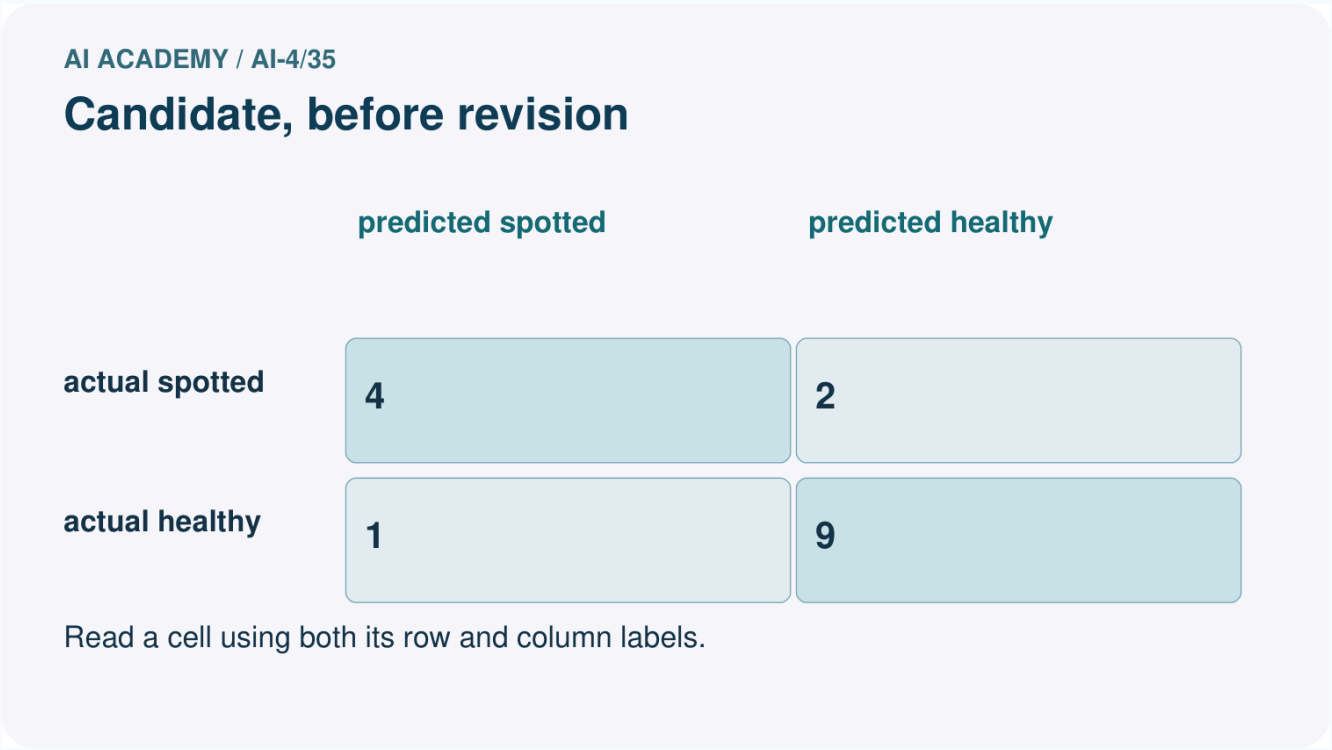
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Train, Analyze, Improve

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/35

Capstone Train, Analyze, Improve

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare accuracy

Candidate13/16=81.25%; baseline10/16=62.5%.



2. Compare detection

Candidate catches4/6 spotted leaves; baseline catches0/6.



3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

CHANGE ONE THING

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair three plausible capstone overclaims

> Name the split and version that actually support a statement.

Complete claims and repairs

Repair “V2 gets 85%, so final performance is85%”:17/20 is inspected validation performance only. Repair “V2 memorizes all training rows, so it must fail”:40/40 training success alone does not settle generalization; compare the separate validation and later frozen final records.

Repair “a post-final fix keeps an untouched test”:once final errors guide the change, those cases are development evidence for the new version. Keep the original frozen results and name the revision.

Explain the remaining limits

- Submit the frozen test report, **error audit**, revision proposal and updated model card. A revision proposal may conclude that more appropriate data or a simpler baseline is preferable.

  If no fresh test is available, stop at a transparent proposal instead of claiming the revision has been independently validated.

What must remain visible when version 2 is proposed without a fresh test?

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A higher overall score proves every important aspect improved. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE A higher overall score proves every important aspect improved.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Explain the comparison with less help

> Keep the full record rather than only its highest percentage.

Current task and response

- Using the complete parcel kit, explain the fixed learner and the single V1-to-V2 feature change. Reconstruct the validation matrices and all gained/regressed case IDs. Choose using the predeclared rule, then state what must be saved before final labels are released.

> **My capstone comparison, freeze and evidence reasoning**

Version later revisions honestly

If dim examples account for both misses, record that **observation** and inspect collection conditions. It is a hypothesis for investigation, not proof that lighting caused the failures.

A proposed feature or model change creates version 2. Scores on the same inspected test must be labelled post-test development.

Pictured error trade-off question

- Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent matrix

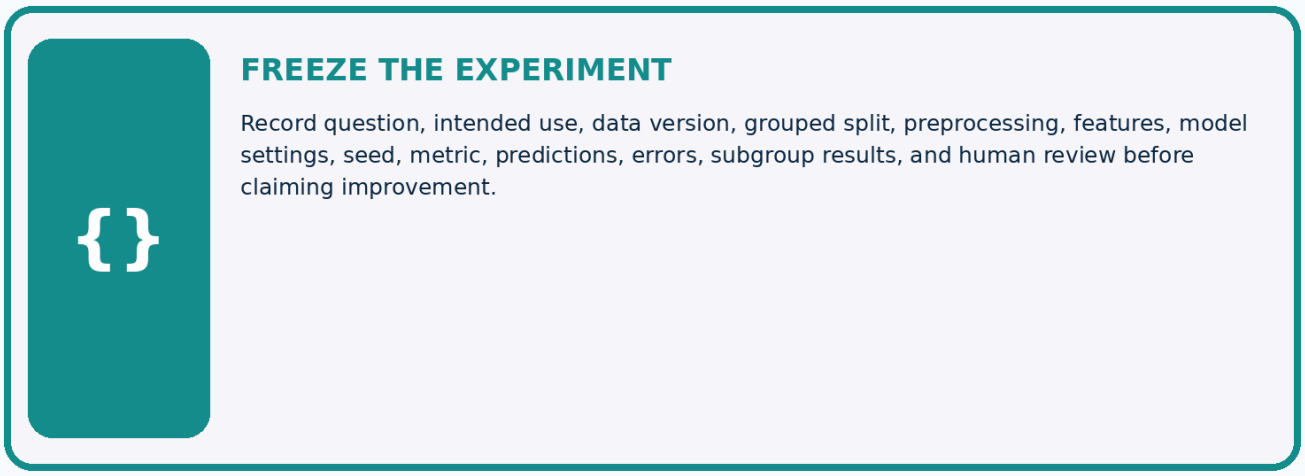
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Audit a difference

Use workbook W2 for the connected example “Capstone evaluation and revision”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

MAIN LESSON · SECTION 9

Run the complete fit,compare and freeze investigation

> The program contains every required record and stops short of final scoring.

Complete runnable student kit and comparison

```
import csv
from io import StringIO
from collections import Counter
```

```
# Complete student kit; final reference cells are deliberately blank.
DATA = """id,group_id,split,width_cm,height_cm,label
P00,P00,training,2,2,clear
P01,P01,training,3,2,clear
P02,P02,validation,4,2,clear
P03,P03,final,5,2,
P04,P04,training,6,2,clear
P05,P05,training,7,2,clear
P06,P06,validation,8,2,clear
P07,P07,final,9,2,
P08,P08,training,10,2,blocked
P09,P09,training,2,3,clear
P10,P10,validation,3,3,clear
P11,P11,training,4,3,clear
P12,P12,training,5,3,clear
P13,P13,validation,6,3,clear
P14,P14,final,7,3,
P15,P15,training,8,3,clear
P16,P16,training,9,3,blocked
P17,P17,validation,10,3,blocked
P18,P18,training,2,4,clear
P19,P19,training,3,4,clear
P20,P20,validation,4,4,clear
P21,P21,final,5,4,
P22,P22,training,6,4,clear
P23,P23,training,7,4,clear
P24,P24,validation,8,4,blocked
P25,P25,final,9,4,
P26,P26,training,10,4,blocked
P27,P27,training,2,5,clear
P28,P28,validation,3,5,clear
P29,P29,training,4,5,clear
P30,P30,training,5,5,clear
P31,P31,validation,6,5,clear
P32,P32,final,7,5,
P33,P33,training,8,5,blocked
P34,P34,training,9,5,blocked
P35,P35,validation,10,5,blocked
P36,P36,training,2,6,clear
P37,P37,training,3,6,clear
P38,P38,validation,4,6,clear
P39,P39,final,5,6,
P40,P40,training,6,6,blocked
P41,P41,training,7,6,blocked
P42,P42,validation,8,6,blocked
P43,P43,final,9,6,
P44,P44,training,10,6,blocked
P45,P45,training,2,7,clear
```

```
P46,P46,validation,3,7,clear
P47,P47,training,4,7,clear
P48,P48,training,5,7,blocked
P49,P49,validation,6,7,blocked
P50,P50,final,7,7,
P51,P51,training,8,7,blocked
P52,P52,training,9,7,blocked
P53,P53,validation,10,7,blocked
P54,P54,training,2,8,clear
P55,P55,training,3,8,clear
P56,P56,validation,4,8,blocked
P57,P57,final,5,8,
P58,P58,training,6,8,blocked
P59,P59,training,7,8,blocked
P60,P60,validation,8,8,blocked
P61,P61,final,9,8,
P62,P62,training,10,8,blocked
P63,P63,training,2,9,clear
P64,P64,validation,3,9,blocked
P65,P65,training,4,9,blocked
P66,P66,training,5,9,blocked
P67,P67,validation,6,9,blocked
P68,P68,final,7,9,
P69,P69,training,8,9,blocked
P70,P70,training,9,9,blocked
P71,P71,validation,10,9,blocked
"""
records = list(csv.DictReader(StringIO(DATA)))
for row in records:
    row['width_cm'] = int(row['width_cm'])
    row['height_cm'] = int(row['height_cm'])
train = [r for r in records if r['split'] == 'training']
valid = [r for r in records if r['split'] == 'validation']
final_features = [r for r in records if r['split'] == 'final']
assert len({r['id'] for r in records}) == len(records) == 72
sets = [{r['group_id'] for r in part} for part in [train, valid, final_features]]
assert not (sets[0] & sets[1] or sets[0] & sets[2] or sets[1] & sets[2])
assert all(not r['label'] for r in final_features)
counts = Counter(r['label'] for r in train)
constant = 'blocked' if counts['blocked'] > counts['clear'] else 'clear'
predictions = [constant] * len(valid)
correct = sum(p == r['label'] for p, r in zip(predictions, valid))
positives = sum(r['label'] == 'blocked' for r in valid)
found = sum(p == r['label'] == 'blocked' for p, r in zip(predictions, valid))
print('roles', len(train), len(valid), len(final_features))
print('training clear/blocked', counts['clear'], counts['blocked'])
print('frozen baseline', constant)
print('validation correct/total', correct, len(valid))
```

```
print('validation blocked found/total', found, positives)
print('final references absent: not scored')
# IDs/groups are audit metadata, not model features. Tie rule chooses clear.

from fractions import Fraction

# Declare the two candidates and selection rule before inspecting validation.
# V2 changes only the available feature set. k=1 and all tie rules stay fixed.
def fit_neighbour(training, features):
    if not training or not features:
        raise ValueError('Need training records and named features')
    if any(r['label'] not in ('clear', 'blocked') for r in training):
        raise ValueError('Training labels must be known')
    lows = tuple(min(r[f] for r in training) for f in features)
    spans = tuple(max(r[f] for r in training) - lo for f, lo in zip(features, lows))
    if any(span == 0 for span in spans):
        raise ValueError('This experiment requires nonconstant training features')
    def scale(row):
        return tuple(Fraction(row[f] - lo, span) for f, lo, span in zip(features, lows,
spans))
    stored = [(r['id'], scale(r), r['label']) for r in training]
    return features, lows, spans, stored

def predict(row, model):
    features, lows, spans, stored = model
    point = tuple(Fraction(row[f] - lo, span) for f, lo, span in zip(features, lows,
spans))
    # Manhattan distance; exact ties use the smallest training ID.
    nearest = min(stored, key=lambda item: (sum(abs(a-b) for a,b in zip(point,item[1])),
item[0]))
    return nearest[2]

def matrix(part, preds):
    if len(part) != len(preds) or not part:
        raise ValueError('Need one prediction per labelled case')
    cells = [0, 0, 0, 0]  # TN, FP, FN, TP; blocked positive
    for r, p in zip(part, preds):
        if r['label'] not in ('clear', 'blocked') or p not in ('clear', 'blocked'):
            raise ValueError('Unknown reference or prediction; do not score blank final
labels')
        cells[2 * (r['label'] == 'blocked') + (p == 'blocked')] += 1
    return cells

models = {'V1': fit_neighbour(train, ('width_cm',)),
          'V2': fit_neighbour(train, ('width_cm', 'height_cm'))}
validation_predictions = {v: [predict(r,m) for r in valid] for v,m in models.items()}
validation_matrices = {v: matrix(valid, ps) for v,ps in validation_predictions.items()}
# Predeclared: most validation correct; then fewer features; then version name.
```

```
chosen = min(models, key=lambda v: (-(validation_matrices[v][0]+validation_matrices[v]
[3]), len(models[v][0]), v))
print('validation baseline TN FP FN TP', matrix(valid, [constant]*len(valid)))
for v in models:
    train_preds = [predict(r,models[v]) for r in train]
    print(v, 'training correct/40', sum(p==r['label'] for p,r in zip(train_preds,train)))
    print(v, 'validation TN FP FN TP', validation_matrices[v])
    print(v, 'validation errors', [r['id'] for r,p in zip(valid,validation_predictions[v])
if r['label']!=p])
gains = [r['id'] for r,a,b in
zip(valid,validation_predictions['V1'],validation_predictions['V2']) if a!=r['label'] and
b==r['label']]
regressions = [r['id'] for r,a,b in
zip(valid,validation_predictions['V1'],validation_predictions['V2']) if a==r['label'] and
b!=r['label']]
print('V2 gains', gains, 'regressions', regressions)
print('frozen choice', chosen)
# Save this code/data/feature contract and these predictions before key release.
final_predictions = [predict(r,models[chosen]) for r in final_features]
print('frozen final prediction ledger', list(zip([r['id'] for r in final_features],
final_predictions)))
print('Final score withheld: teacher reference release and freeze check required.')
```

Freeze checklist and release boundary

Freeze field	Complete kit contract / action
Data and groups	72-row kit:40/20/12, one parcel per group; retain exact membership and CSV/code copy.
Model and preprocessing	V2, features width_cm/height_cm, training lows 2/2 and ranges 8/7, k1, Manhattan, exact Fraction arithmetic, smallest-ID tie.
Selection evidence	Same20 validation rows, V1 13/20, V2 17/20; four gains, no regressions, three remaining misses.
Frozen predictions	Program prints one saved prediction for each of12 reserved feature IDs without reading a final reference.
Teacher gate	Teacher checks saved record and records actual date/status before key release; no completed approval is invented here.
After release	Score frozen baseline and candidate on identical final IDs; save every result. Any changed version gets a new name and reused-evidence label.

Normal,tie,blank-label,constant-feature and target-leakage probes

Complete program probe	Expected check
Normal fixed comparison	V1 13/20, V2 17/20; choose V2.
Exact equal distances	Smallest training ID wins; do not choose using the validation label.
Blank final references	Print predictions only; matrix() rejects scoring blank labels.
Constant training feature	Fit raises ValueError under this stated nonconstant-feature experiment contract.
Changed query reference only	Prediction unchanged:predict reads feature fields, not the query label.

Run or trace and retain the actual record

1. Capstone Gate 2: predict the comparison, run or trace the complete code, then retain the actual output and code copy. Check all gains and remaining errors. Save the freeze fields before the teacher releases references. Running this supplied paper kit reproduces a constructed example; it does not perform a real classroom review or evaluate your own plant project.

   Apply the same record structure to your actual project without copying these numbers as its results.

Keep the experiment ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Capstone Gate 2: reproduce v1, cluster errors, approve one change, rerun fixed cases, and publish all gains and regressions.

1. Write the hypothesis and expected **evidence** before running.

2. Capstone Gate 2: reproduce v1, cluster errors, approve one change, rerun fixed cases, and publish all gains and **regression**s.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit a fresh frozen evaluation

> Explain accuracy,precision,recall and error trade-offs from the supplied matrix.

Complete selected matrix and axes

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

This is a separate fully supplied 12-case frozen-matrix exercise, not the parcel kit's hidden result. Rows are true clear/blocked and columns predicted clear/blocked; blocked is positive. Use its given counts and compare both kinds of error. Keep the first attempt before opening the teacher key; do not substitute the worked validation or pictured 16-case numbers.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A revision raises overall validation accuracy from **80%** to 84% but lowers dim-light accuracy from **60%** to 40%. How should the improvement claim change?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Build an inspectable capstone record

> The reader should be able to separate selection evidence from final evidence.

Project artifact and complete record

THIS WEEK'S CHECKPOINT Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

Keep the complete hypothesis, V1/V2 code and data versions, split/group membership, training-only preprocessing, all validation predictions/gains/regressions and actual freeze status. Your Plant Classifier project may have different labels, inputs and data; the parcel-kit scores are not its evidence.

After final release preserve baseline/candidate comparisons unchanged, and label any error-driven revision as a new development version needing fresh final evidence.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Scope,human decision and remaining limits

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

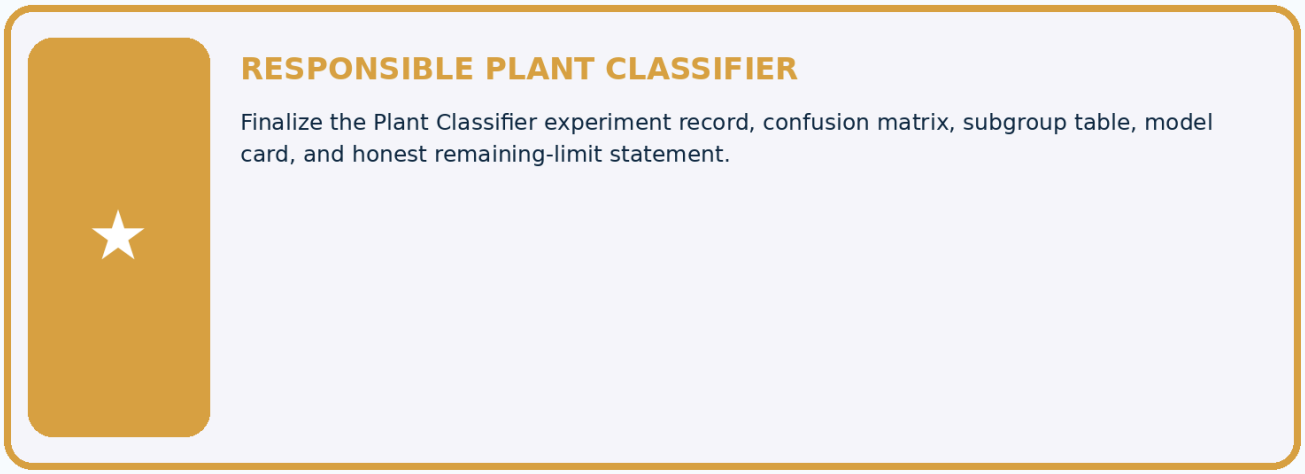
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with a selected version and a bounded claim

> Name one remaining failure and the evidence needed next.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Run a frozen **final evaluation** with complete accounting

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
baseline run	first reproducible model result	
error audit	systematic review of mistakes	
before/after	same evaluation around one change	
regression	previously passing case now fails	

Closer-explanation terms

Word	Meaning
Final evaluation	Assessment of the frozen procedure on held-out evidence
Error audit	A documented inspection of mistakes and missing predictions
Revision log	A record of changes, reasons and affected evidence

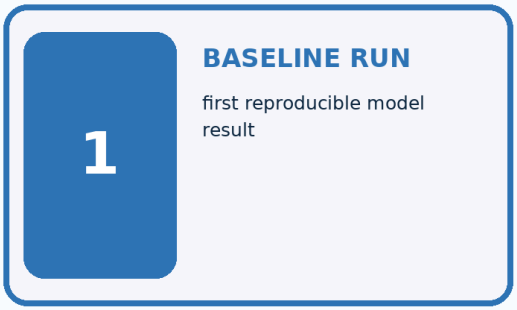
Word	Meaning
Evidence status	Whether a result is development, untouched test or post-test reuse

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A higher overall score proves every important aspect improved.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **baseline run**, **error audit**, **before/after**, **regression**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A higher overall score proves every important aspect improved.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

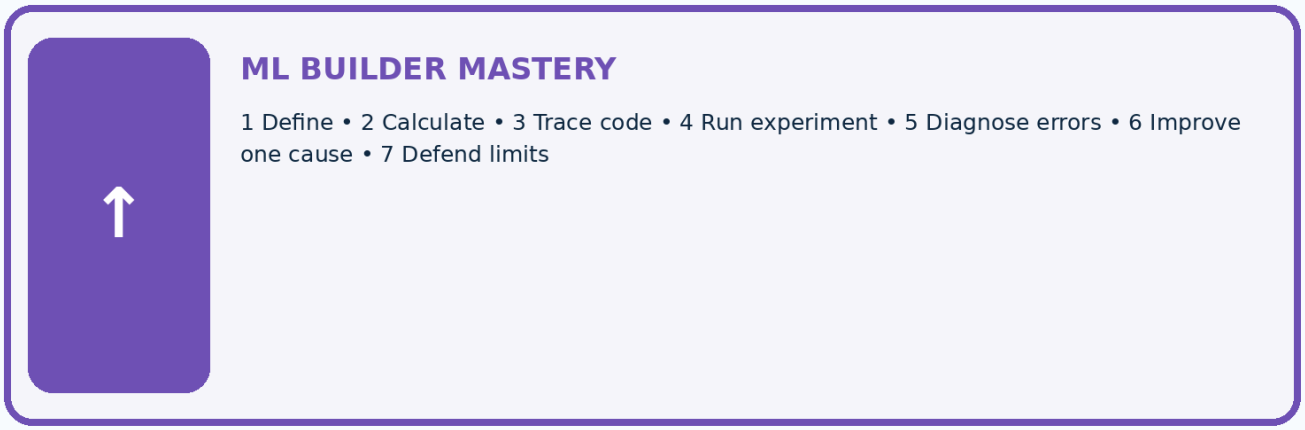
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected separate12-case frozen matrix for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

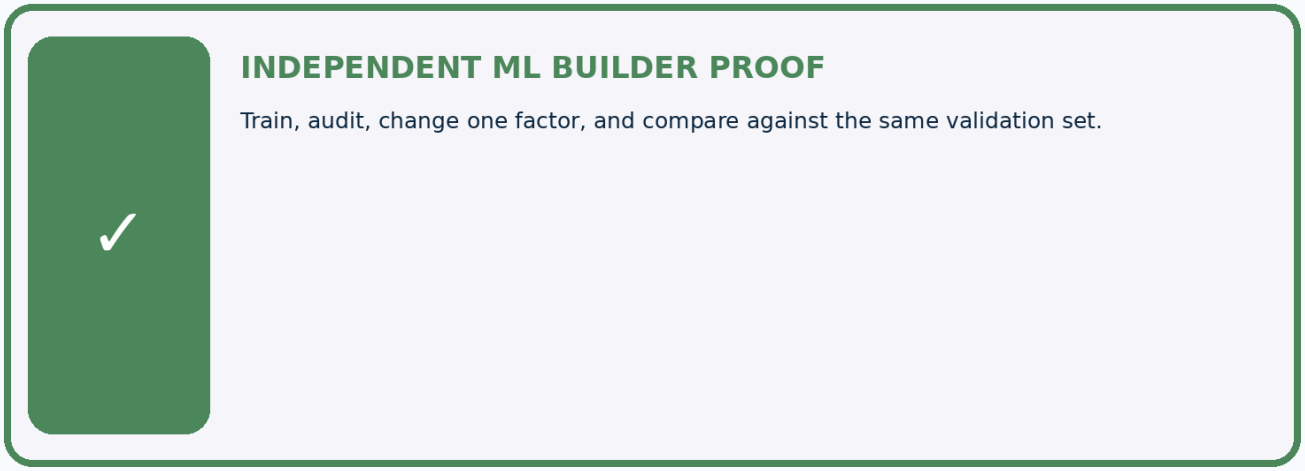
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Train, audit, change one factor, and compare against the same validation set.

NEXT BRIDGE Carry the experiment **evidence** into: ML Builder Expo.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Plan improvement **evidence**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

Evidence or stage	Value or result
Baseline	Seventy percent validation accuracy
Model	Eighty percent on the same set
Error slice	Many failures in dim images
Next revision	Change one factor and retain final test

▼ Earlier wording, alternatives and picture transcripts

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Keep improvement evidence comparable

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Plan improvement **evidence**.

Keep partitions fixed, inspect dim-image errors, change one factor and compare the same validation criteria/slices. Reserve final test.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Run a frozen final evaluation with complete accounting	Explain the deciding step, show a trace or test, and state one limit.
Separate error analysis from an untouched-test claim	Explain the deciding step, show a trace or test, and state one limit.
Document revisions and their evidence status	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Choose the next capstone revision from explicit false alarms and misses.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Choose the next capstone revision from explicit false alarms and misses.

- Which labelled step matches this detail? Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising.
- 1. Inspect errors
- 2. Compare accuracy
- 3. Compare detection
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Inspect errors Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. Read the picture in order.

Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A revision fixes one miss but creates two extra false alarms. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Repair and keep the record

Review Capstone Dataset and Baseline (ai-4/34). Goal: Define a feasible capstone question and acceptance criteria. Short practice: What provenance field prevents the crop from appearing independent?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

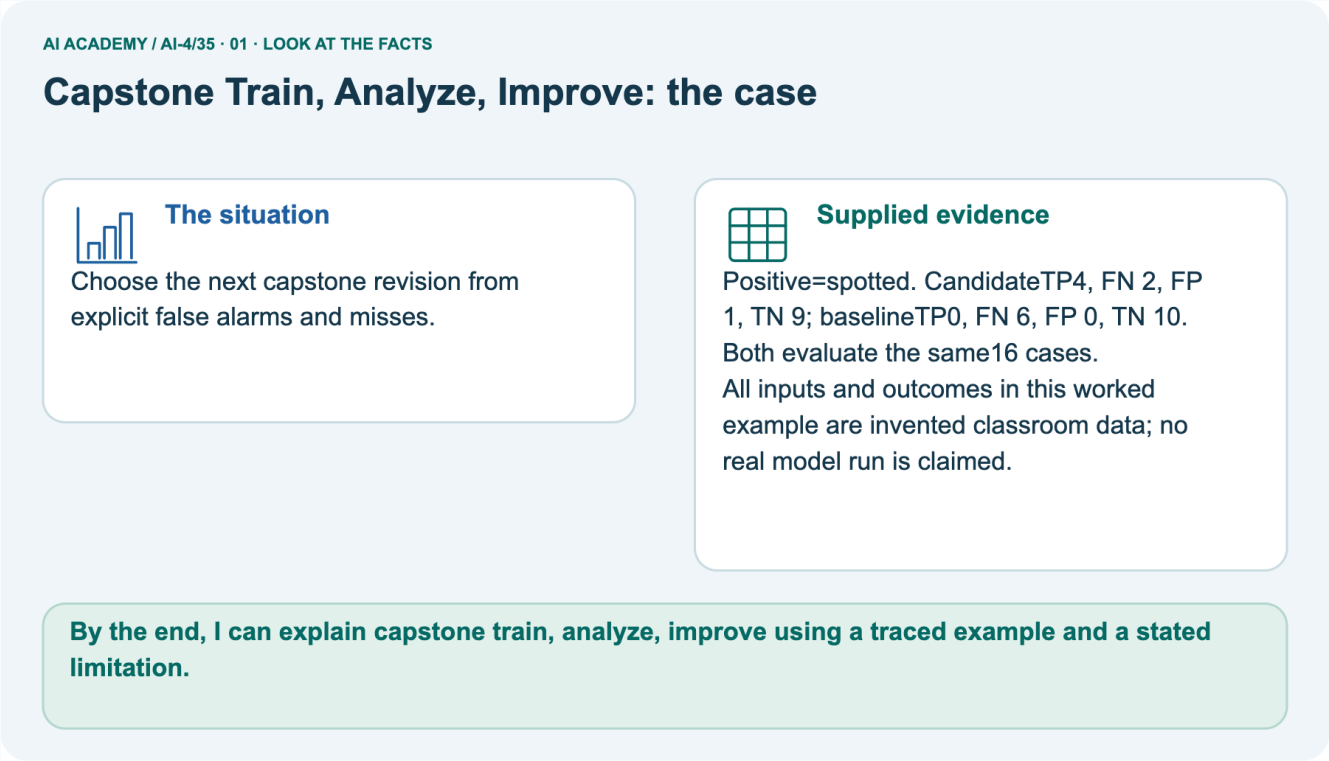
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Capstone Train, Analyze, Improve: the case



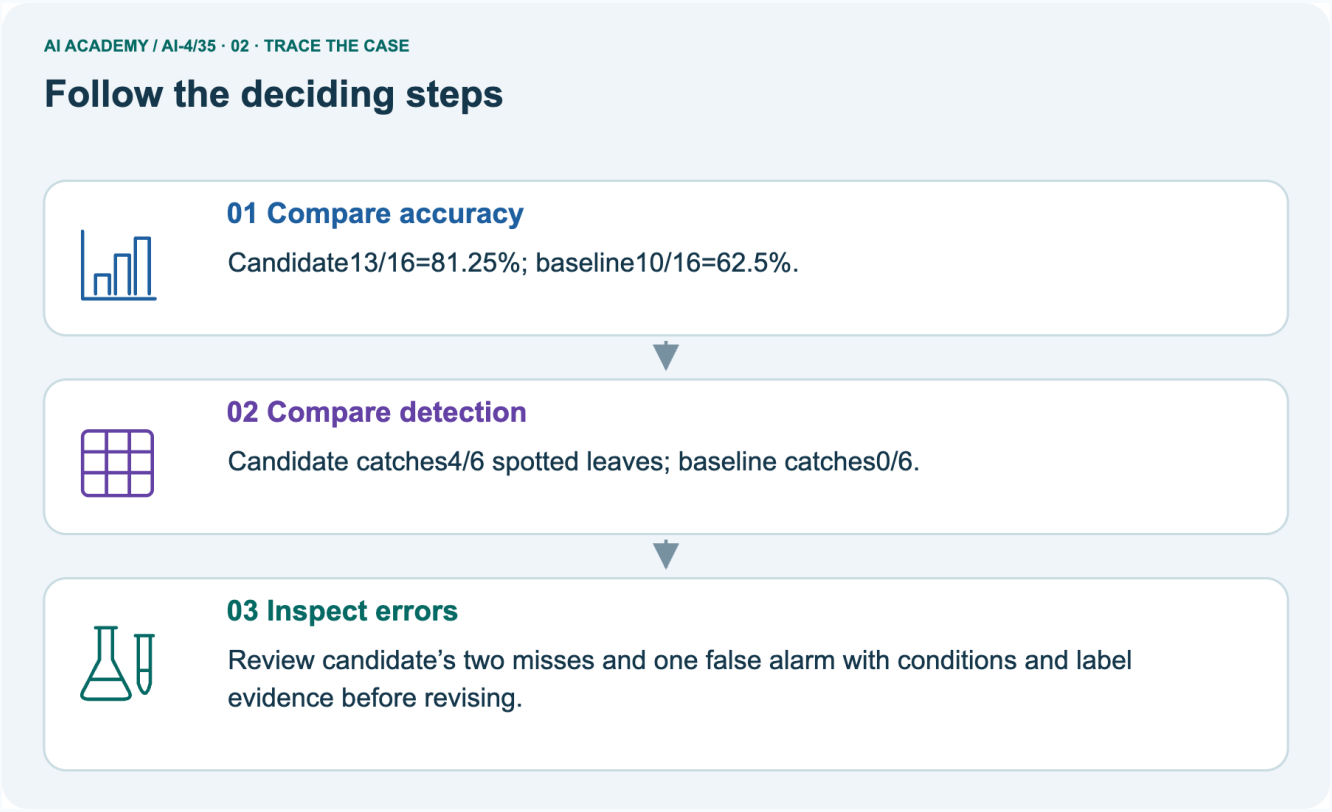
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose the next capstone revision from explicit false alarms and misses. Supplied facts: Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

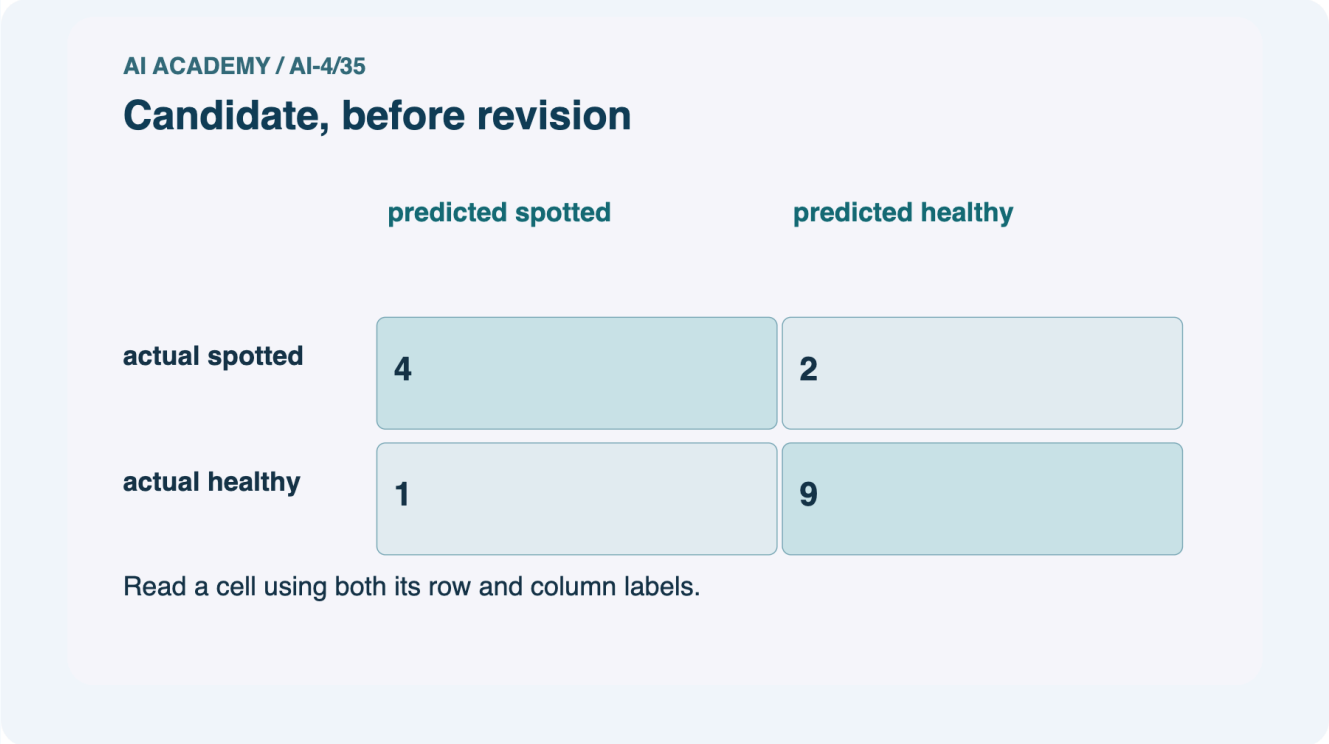


Follow the labelled picture with the matching explanation. Open the vector diagram

Compare accuracy: Candidate13/16=81.25%; baseline10/16=62.5%. → Compare detection: Candidate catches4/6 spotted leaves; baseline catches0/6. → Inspect errors: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

Point to each numbered stage and explain which recorded **input** it uses.

03. Candidate, before revision

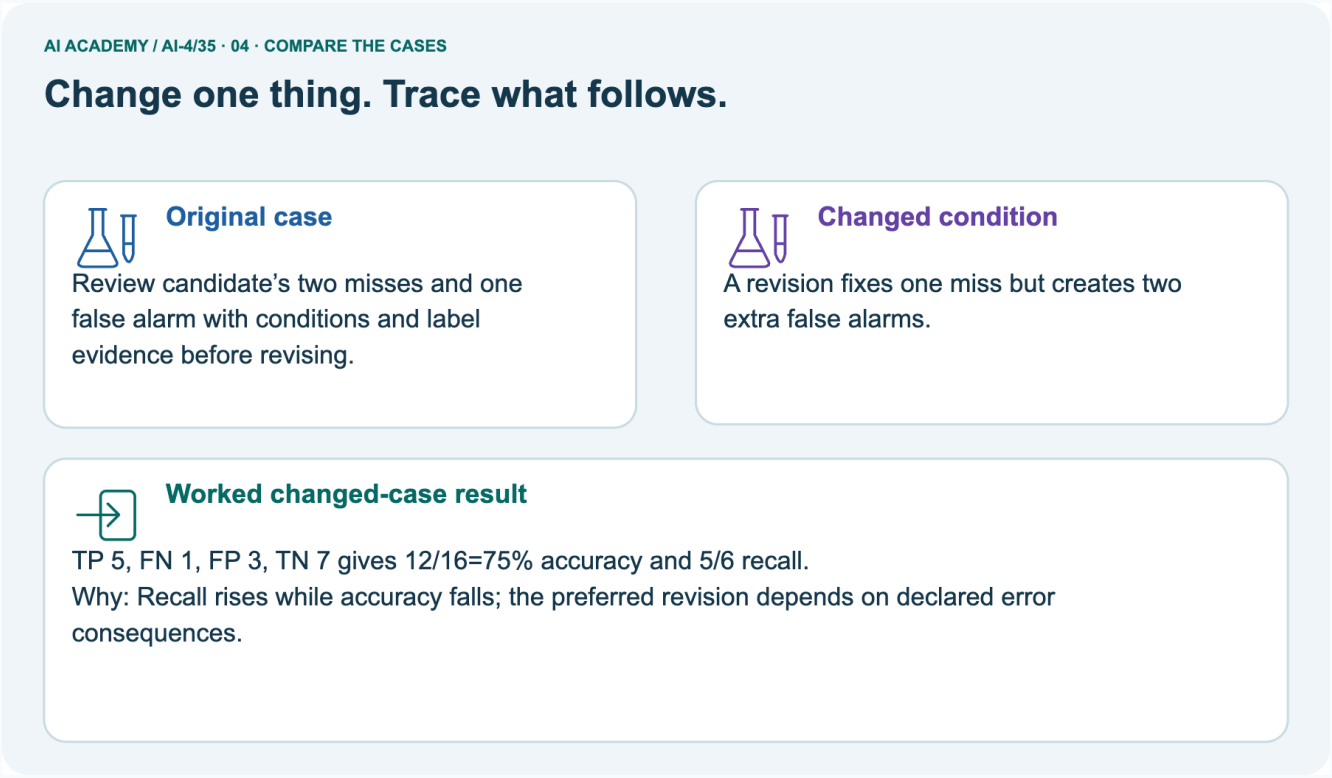


Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate, before revision. actual spotted: predicted spotted = 4, predicted healthy = 2. actual healthy: predicted spotted = 1, predicted healthy = 9.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



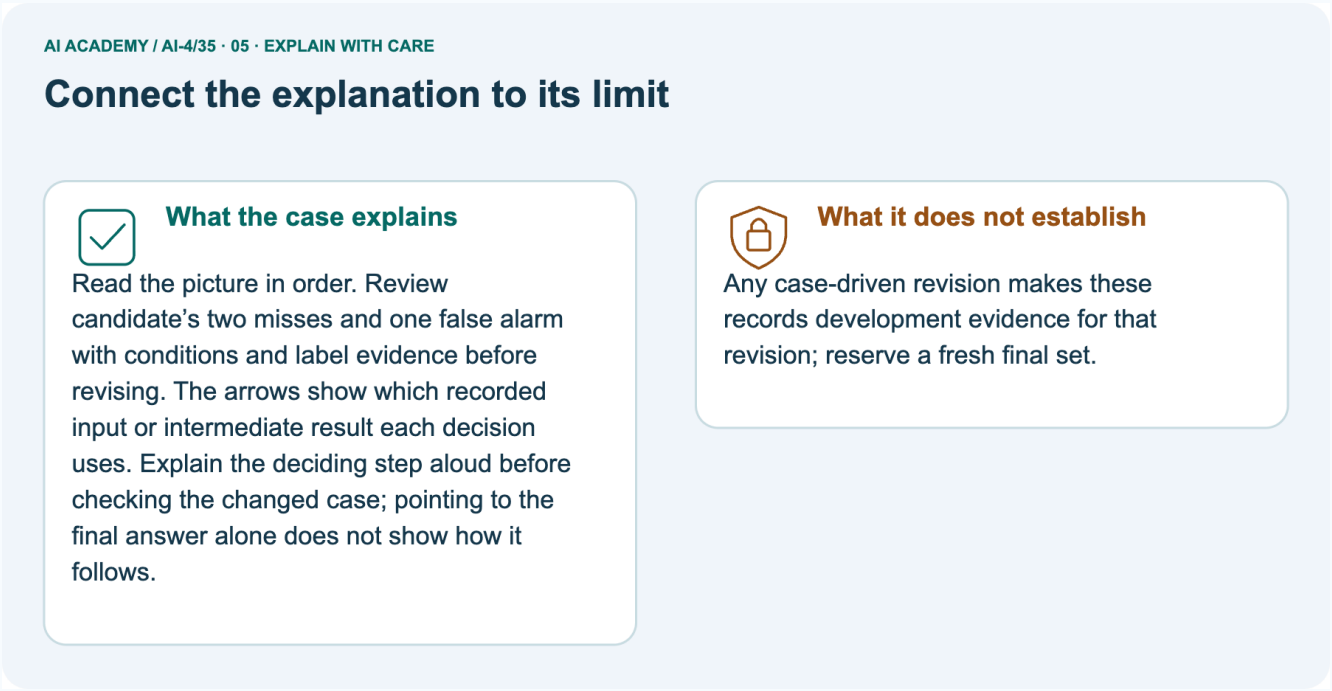
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Changed condition: A revision fixes one miss but creates two extra false alarms.

Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

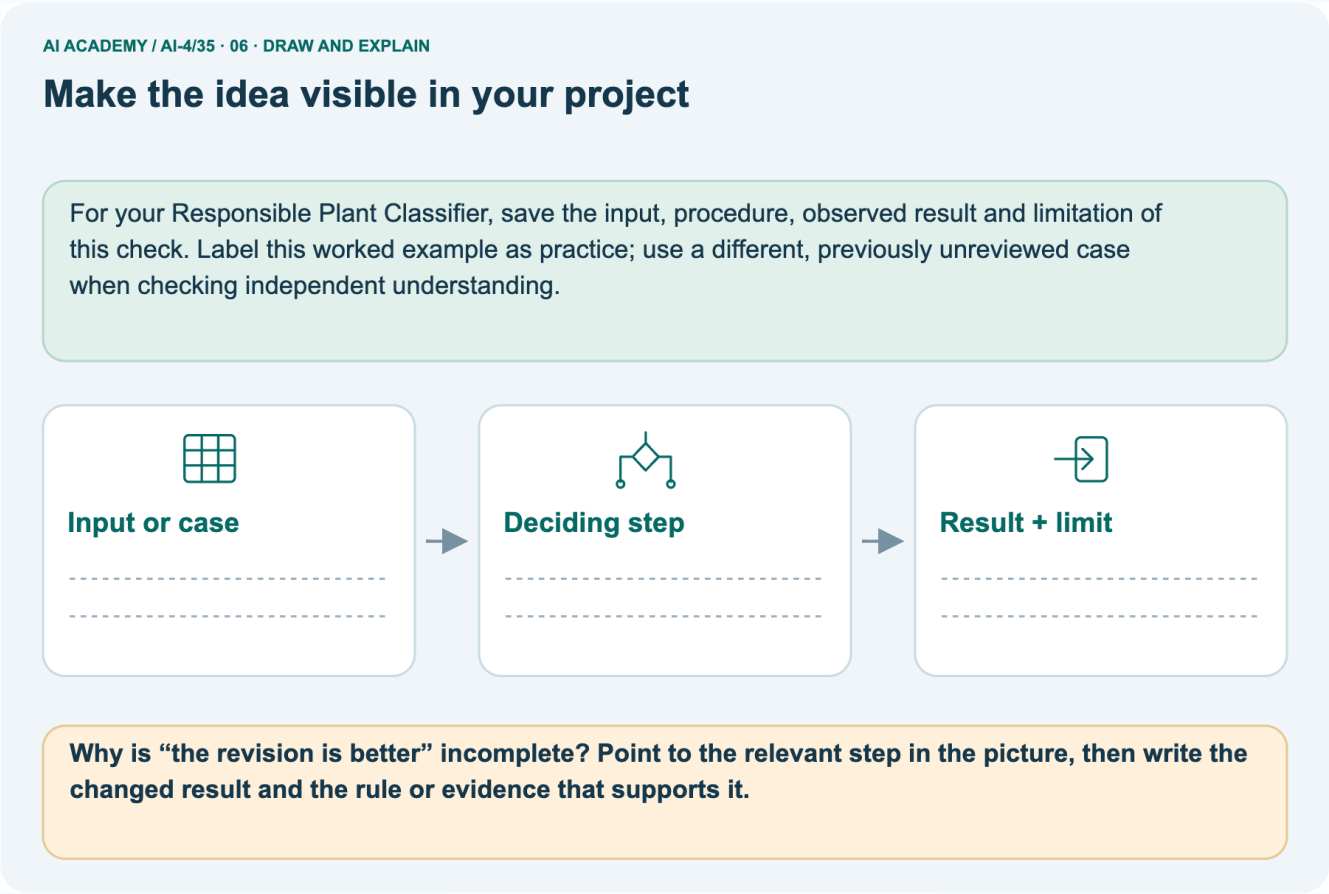
Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Capstone Train, Analyze, Improve: the case

02. Follow the deciding steps

03. Candidate, before revision

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.